

Predicting any observed quantity of the Earth:
a design for a global ocean–land–atmosphere model
that reads the observational record through a learnable space–time cone

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Abstract

We specify a two-stage model that predicts any observed quantity of the Earth system — a grid cell, a station, a float profile, a satellite pixel — at any place, time and footprint, from the whole observational record of ocean, land and atmosphere. The design rests on one sampling prior: a quantity at a place and time is most informed by a cone in space–time whose reach grows with the speed of the fastest signal carrying information about it and ends with that signal’s memory, and it informs the mirror of that cone into the future. Both stages sample this hourglass, supplemented by an ungated global far field. Stage 1 reads raw values over the last thirty days through a waist of several cells and writes one embedding per cell and pentad. Stage 2 reads those embeddings over two years at log-spaced lags and writes the embeddings of the next year as a residual on an asymptotically stable linear propagator fitted as a linear inverse model. Every value enters through one token — value, observed flag, offset from the anchor, footprint, source — so dense 0.25° grids, kilometre-scale fields, tile catalogues and point observations share one encoder without resampling to a common analysis grid; every prediction leaves through one query-based decoder that takes the same coordinates and footprint. The cone’s shape — a drift toward the source, a waist ellipse, a growth ellipse, a scale and a memory — is learned per 1° cell and per channel; what the model reads is learned, what it is asked to predict is drawn from a fixed distribution that the learned geometry cannot move. The first experiments address initialised forecasts of observed Earth-system quantities over five days to one year; long-term forced responses and scenario projections are outside the scope and remain unvalidated. We state the data, the geometry, both stages, the training ladder, the evaluation contract — which periods are development, retrospective evaluation and prospective test, and what each source was available to know at issue time — the null ladder and the comparisons that decide each choice, beginning with a fixed-target forecasting benchmark that asks whether the representation improves a forecast beyond cheap models given the same information. Verdict rules are fixed before any run. A results section is reserved with the tables it will hold.

1 Introduction

We specify a model that, given the observational record up to time t , returns the value an instrument would read at place $\mathbf{x}$, time $t+\ell$, depth or height, and footprint (the area and span the reading averages), for any channel the record has contained. A transport through a mooring section, an El Niño index, a coastal pixel’s chlorophyll, a station’s temperature, a gauge’s sea level, a forest’s canopy height at a laser footprint and a tower’s carbon flux are instances of one query, answered by one representation and one decoder. The Atlantic overturning, the ocean carbon sink, El Niño, sea-surface temperature and the land biosphere are the five read-outs tracked, because each has a long truth series; they are properties of the instruments, not of the target.

The design rests on one sampling prior, stated in §2.1: what bears most on a value at a place and time lies in a cone in space–time, and the value bears on the mirror of that cone (Figure 1). The prior is transport-inspired and is not a claim about causation; the model also reads an ungated far field, and what its learned geometry represents is predictive relevance. The record at 0.25° and five-day cadence is of order 10^9 values a year, the kilometre tier a thousand times more, and point observations are irregular in every coordinate; no transformer attends over that directly. Stage 1, the codec, reads raw values in a small cone — thirty days, hundreds to a few thousand kilometres — and writes one embedding per cell and pentad; it is the only stage that sees a raw value. Stage 2, the forecaster, reads embeddings in a large cone — 35 days to two years, up to the planetary scale — and writes the embeddings of the

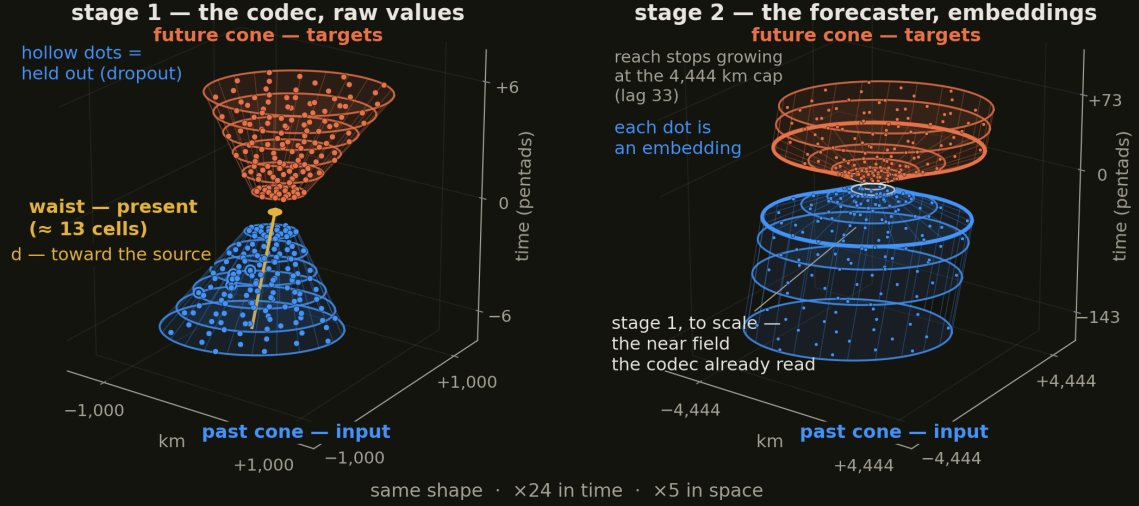


Figure 1: The hourglass at both scales. Each lag is a 24-point sunflower (phase-rotated by the golden angle per lag) inside an ellipse that grows with lag; past half input, waist present, future half target only. Left: stage 1 over raw values, lags 1–6 and leads 1–6 pentads, ≈ 900 km at six pentads for the ocean surface, a thirteen-cell waist. Right: stage 2 over embeddings: past slices at lags 7, 13, 21, 34, 55, 89 and 143 pentads (the lag-0 embedding is the waist), future targets at leads up to 73 pentads, reach to a 4,444 km cap; the drawing shows the slices the text lists (§2.6), and stage 1’s hourglass is the small solid at the centre.

future cone; it is what is rolled forward and scored. Stage 1’s hourglass is the small solid at the centre of stage 2’s. The split is in time, not distance.

Premises. (i) At one basin’s data volume the forecast is insensitive to width and its held-out optimum arrives early. That is consistent with a data limit and also with a limit in the representation, the objective, the optimisation or the target noise; the design grows data before parameters and measures which it is (§4), each rung of the ladder being bought on the one below. (ii) A linear inverse model and damped persistence are strong beyond two weeks, so the learned forecaster is a residual on a linear propagator fitted as a linear inverse model and is reported against both. (iii) A read-out with few effective degrees of freedom cannot resolve a design choice; comparisons are decided on the dense field, with a fixed panel of decoded physical quantities beside it. (iv) The evaluation contract separates development years, a retrospective evaluation block that has informed design decisions and is labelled as such, and a prospective period that no statistic, model or decision has touched; the exclusion of held-out bins from every window the forward pass can see is certified before training, and every source carries the time at which it became available. A technical report on the North Atlantic pilot [1] is the dated record behind these premises; this paper does not repeat it.

Scope. The first experiments address initialised forecasts of observed Earth-system quantities over five days to one year from the observational record up to the issue time. Multi-year rollout is a mode of the forecaster, not its objective, and nothing in this design supports a claim about scenario-conditioned change, tipping times or century-scale behaviour; a scenario-conditioned projection needs a forcing pathway as an input and an evaluation of the response under changed forcing, which a longer rollout of an observation-conditioned model is not. The query interface is broad; demonstrated predictability is whatever §6 will show.

Section 2 states the method: the problem, the token every observation becomes, the cone and its learnable geometry, the two stages and the query decoder. Section 3 describes the data, Section 4 the training ladder, and Section 5 the evaluation protocol and the comparisons that decide each design choice, with verdict rules fixed before any run. Section 6 reserves the results; Section 7 places the design among existing models.

2 Method

2.1 Problem formulation

Queries. Let $\mathcal{O}_{\leq t}$ be every observation made up to time t : gridded fields, point and profile measurements, image tiles, each a value with a place, a time, a depth or height and a footprint. A query $q = (\mathbf{x}, t + \ell, z, c, f_p, f_t)$ names a place, a lead $\ell \geq 0$, a depth or height, a channel c and a footprint in space and time. The model returns a predictive distribution $p(y_q \mid \mathcal{O}_{\leq t})$ — a mean and a variance in the channel’s transformed space (§2.5) — for any query whose channel and footprint occur among the training targets. Lead zero with the value hidden is a nowcast. Every stage below serves this one function.

The dependency cone as a sampling prior. A driver moving at speed v reaches $\mathbf{x}$ at time t from within $v \Delta t$ at $t - \Delta t$. The points that can carry a signal to $(\mathbf{x}, t)$ form a cone opening into the past, its reach set by the fastest mechanism carrying that channel and cut off by the channel’s memory: wind stress forgets in days, sea-surface temperature anomalies in months, a deep water mass in years. Each channel has a family (v, τ) . Run forward, the same argument gives the points $(\mathbf{x}, t)$ can carry a signal to: the point mirror of the past cone. Past cone, waist and future cone are an hourglass (Figure 1), and advection is what the mirror looks like: water came from upstream and goes downstream. This is a prior on where to sample, not a law the model recovers. Predictive relevance is not causal influence: a remote observation can reveal a shared driver or a hidden state without any parcel travelling to the target, coupled pathways and spatially varying flow break the one-channel-one-drift picture, and a dissipative forward system does not imply a point-mirrored statistical dependency. So the design tests a transport-inspired space–time sampling prior, supplemented by a global far field; the geometry it learns represents where the predictive information was found and is read as a diagnostic, never as a recovered transport law.

2.2 One token per observation

The encoder attends over places at times, each carrying every channel there:

$$\text{token} = (\mathbf{v}[C], \mathbf{m}[C], \Delta x, \Delta y, \Delta z, \Delta t, \log_2 f_p, \log_2 f_t, n_R, s, \text{sphere}, \text{season}). \quad (1)$$

$\mathbf{v}$: standardised anomalies of the group’s channels; $\mathbf{m}$: observed flags; offsets from the anchor in km, m and days; footprint $\log_2 f_p = \log_2(\text{km}/27.83)$ and $\log_2 f_t = \log_2(\text{days}/5)$, clamped to ± 12 (a Sentinel-2 pixel $(-11.4, -4)$, a laser footprint $(-10.1, -12)$, a 30 m annual map pixel $(-9.9, +6.2)$, an Argo profile $(-4, -4)$, a 4 km daily cell $(-2.8, -2.3)$, an OISST pentad cell $(0, 0)$, a reanalysis cell at its true support $(+2.9, 0)$, a monthly 1° Argo field $(+2, +2.6)$); n_R : how many such observations were within reach; s : a learned source embedding; sphere code; day-of-year harmonics. One token per location and lag, not per channel: adding a channel adds a vector entry, not a token. Footprint is constant per group for dense sources and per row for sparse ones; nothing else depends on the storage tier (§3.2), so the encoder never learns which file a value came from. The two footprint scalars are a first approximation: they encode the support’s scale, not its shape, and a swath cell, a laser footprint and a monthly objective analysis have supports a scalar cannot describe. Each source therefore carries its documented support assumption in the registry (§3.2), and consistency between a fine and a coarse reading of the same quantity is tested only where the coarse measurement is an average of the fine one; the decoder is never asked to invent fine-scale information from a coordinate alone. A 1° value is one token at the cell centre, seen from sixteen offsets by the sixteen anchors in it. Point observations enter where and when they were made: the k nearest in space–time within the cone, k small (three to eight per depth group), one-sided in time. Three absences are kept apart. A slot the nearest-neighbour search cannot fill is a whole padded token, excluded from attention by the key mask; it is bookkeeping, not evidence. A channel entry the source never measured (a land cell’s SST, a month without Argo) is a learned *never-observed* entry inside an attended token, with its observed flag off. A channel entry the encoder may not see during training is a learned *hidden* entry, distinct from never-observed, so that “this is being withheld” and “this was never measured” are different facts to the model.

2.3 Sampling the space–time cone

Waist. The present is a disc of radius two cells (≈ 13 cells, a small sunflower), read and reconstructed; the embedding z is written for its centre. Gradients need neighbours: thermal wind is a density gradient, geostrophy a height slope, and neither exists in one cell. The waist carries the cone’s shape at lag zero, so a jet can have an elongated waist.

Past cone. Lags 1–6 pentads, every pentad: five days is the tensor’s cadence and thirty days is the codec’s job. At each lag the sampling set is a 24-point Vogel sunflower $\mathbf{u}_i$ mapped through a Gaussian footprint,

$$\Sigma(\ell) = \Sigma_0 + \ell^2 \Sigma_v, \quad p(\ell, i) = \mathbf{x}_{\text{anchor}} + \ell \mathbf{d} + \Sigma(\ell)^{1/2} \mathbf{u}_i, \quad (2)$$

with drift $\mathbf{d}$ per pentad toward the source, waist ellipse Σ_0 and growth ellipse Σ_v (standard deviation linear in lag). The sunflower’s phase advances by the golden angle per lag, so the six slices interleave. The anchor column is sampled at every lag.

Far ring. Every lag also keeps a fixed ring of 24 points at log-spaced radii from 28 km to the antipode, read at full weight and never gated. The design speed is the fastest measured mechanism of the family times 1.5 (the ocean: 3.6 ms^{-1} , so 5.4 ms^{-1} and 2,333 km per pentad; wind stress: the atmosphere, global within a pentad). The ellipse gives direction and density; the ring keeps a sparse, ungated reach to the whole globe at every lag, so that a remote driver is not excluded by the cone’s shape. Twenty-four points are not exhaustive coverage of the far field, and the ring is not a guarantee: a compact coarse global summary of equal token budget is the comparator that tests it (§5.2).

Prediction cone. The past cone point-mirrored: centre $-\ell \mathbf{d}$, same $\Sigma(\ell)$, leads +1 to +6, sampled the same way, read by nobody. Each point is a decoder query and a target. The anchor column at every lead is always a target. The target positions are drawn from the *fixed* hourglass of the initial geometry (§2.4) and are shared across arms and seeds: the model learns where to read, and what it is asked to predict does not move.

2.4 Learnable per-channel geometry

Per-channel geometry. What moves a signal differs by channel even inside one file group: currents follow the flow; sea-surface height anomalies move west with eddies and Rossby waves whatever the current does [3]; sea ice moves with the wind at 1–2% of its speed and to its right [4]; sea-surface temperature follows the wind at short lags and the current at long ones [5]; below ~ 800 m at the western boundary the deep current runs opposite to the upper limb [6]. So every geometry number is per channel: $\mathbf{d}_c$, $\Sigma_{0,c}$, $\Sigma_{v,c}$, a log-scale s_c and a memory τ_c (two scales for the L-shaped channels, lags 0–1 and 2–6). They are stored hierarchically — a group value plus a per-channel residual under L2 shrinkage — so a channel that shares its group’s carrier costs nothing and one that does not pays for its departure with evidence. Initial values come from a table set by physics: wind stress 500 km with a ten-day memory; currents, height and profiles 130 km per pentad with a long memory; SST and mixed layer $\max(\text{that}, 500 \text{ km})$ at lags ≤ 1 then the ocean’s growth; land 400 km flat, no drift; $\mathbf{d} = 0$ everywhere; the waist ellipse at the measured flow anisotropy 0.71. Step zero is the fixed hourglass. The biosphere channels are land channels: no carrier of their own, an aperture that opens over the atmospheric and soil tokens, fire the one exception with the wind as its carrier at 5–30 km a day. Their memories — weeks for leaf area, decades for canopy height — exceed the cone, so a slow channel is read as a state: the previous annual map is an input token at lag ≥ 73 pentads and next year’s map at the same footprint is the target; the same hourglass at the annual time footprint.

Aperture, at zero extra tokens. A token carries all channels of its group, so each channel reads the shared points through its own weight

$$w_{c,i,\ell} = \exp\left(-\frac{1}{2} (p_i - \ell \mathbf{d}_c)^\top \Sigma_c(\ell)^{-1} (p_i - \ell \mathbf{d}_c)\right) e^{-\ell \Delta t / \tau_c}, \quad (3)$$

applied to that channel’s value and flag entries before projection. A point outside channel c ’s aperture in space or memory contributes ≈ 0 for c and full weight for the others (400 km at lag 1: currents 0.30, SST 0.73 at the initial apertures). The weight is smooth in every geometry number, so the loss reaches

all of them through every point at once. The same weight applies to tier-P tokens, whose offsets and depths are known: each pressure-level channel gets its own direction over the same k nearest profiles, with no sampling ellipse for profiles. The ring is not gated.

Parameterisation. $\mathbf{d} = \tanh(\mathbf{x}) v_{\text{design}} \Delta t$, capped by construction. Ellipses as matrix logarithms, $S = \log \Sigma = m\mathbf{I} + k \begin{pmatrix} \cos 2\theta & \sin 2\theta \\ \sin 2\theta & -\cos 2\theta \end{pmatrix}$: no kilometres, no angle to wrap, a circle is zero, and an Adam step moves every radius by the same fraction of itself [7]. Bounds are softplus offsets in log space, not clips, and they are of three kinds: a *minimum radius* — the waist ellipse is floored at one cell so a gradient always has a neighbour; a *maximum reach* — the drift is capped at $v_{\text{design}} \Delta t$ by the tanh above and the growth ellipse’s standard deviation at lag ℓ is capped so that the cone at the longest lag stays inside the design reach; and a *default growth rate* — the initial Σ_v , 130 km per pentad for the ocean’s carried channels, which is a starting value and not a bound. The design speed is a ceiling on how fast the cone may spread, never a floor on how fast it must. The numbers live on a 1° map per channel (64,800 cells; bilinear blend at the anchor; ≈ 34 M nominal, half live, no compute), because the currents that carry the overturning are ~ 100 km wide and a 10° cell would learn a drift near zero. Each map is a coarse 10° map plus a 1° residual under L2 and Laplacian priors, so a data-poor cell inherits its region. A first-harmonic seasonal drift is a second-phase option for monsoon regimes.

Learning: gate first, harden later. In phase one the candidate points are fixed — a log-radial set of 36 per lag to the design reach plus the anchor column — so reads are regular and independent of live weights, and the geometry learns through the aperture alone. Every 2,000 steps the dense table per region is re-baked, the 24 points per lag drawn from the group’s mixture of channel ellipses, and swapped at an epoch boundary. Geometry parameters get ten times the codec’s learning rate [8]. Reading at continuous positions was rejected: it ties the loader to live weights and its gradient sees four cells.

Guards. A movable question moves to easy water, and a stop-gradient on target positions does not stop the target *distribution* from changing when the targets are re-drawn from a learned cone; nor can a variance statistic detect a move toward cleaner or denser observations. So the input geometry is learned and the target geometry is not: target queries — the mirrored future cone, the hidden waist, the fill-in cells and points — are drawn from the fixed hourglass of the initial table in every phase, including after each re-bake, and the same target set serves every arm and seed of a comparison. The mirror of the *learned* cone is not a training target; whether a learned mirrored target set buys skill is a separate, later comparison with its own guard. The remaining guards are: the aperture never enters the loss weights; the ring stays ungated; the evaluation query set is frozen and identical across arms (§5.1); and the learned geometry is reported as a diagnostic — agreement between a learned drift and a known current is descriptive, since a disagreement can be useful information routing and an agreement can be redundant source structure, and neither establishes a causal reading.

2.5 Stage 1: encoding observations

Tasks. Each example is one anchor at one bin and draws one task (Table 1). Dropout patterns fire independently: one channel hidden at the waist; lags $\leq \ell_0 \in \{1, 2, 3\}$ hidden; a 90° bearing wedge hidden; points hidden at random ($p = 0.3$); for the nowcast, the whole waist. The task is not signalled by a token; hidden markers and the lead in each query suffice. Point observations — a profile, a laser footprint, a fluorescence sounding, a tower’s half-hour — are hidden and predicted like cells; an annual map is a T1 or T4 target at its own footprint and year, with the previous year’s map visible. Three rules are not knobs: in the main tasks a target’s channel must be visible somewhere in the input (the sampler counts violations, required zero), which keeps a copy or persistence control inside every example; the waist family scores hidden channels only; and every (task, family, lead) is reported against the score of predicting the mean, and is called learned only where the ratio is below one held out. The first rule also bars the cross-variable inference a completion model is often wanted for — a subsurface profile from the surface alone, chlorophyll from physics alone — so a fourth, small task, **T4c**, scores exactly that: the whole target channel is absent from the input, on a fixed list of (target channel, visible channels) pairs, with its own predict-the-mean bar. A claim of general observation completion is made only from T4c.

task	input	targets	share	what it teaches
T1 forecast	waist + past cone	future cone	0.35	the full problem
T2 snapshot forecast	waist only	future cone	0.15	what history buys, inside one model
T3 nowcast	past cone under dropout; waist hidden	waist	0.25	the present from the past
T4 fill-in	waist + past cone under dropout	hidden cells and points	0.20	reconstruction from partial observation
T4c cross-variable	waist + past cone with the target channel absent everywhere	that channel, on a fixed pair list	0.05	inference across variables, scored apart

Table 1: Stage-1 tasks, drawn per example. Every target position is drawn from the fixed hourglass. Shares are a starting point and the first thing §5.2 varies.

Model. A masked autoencoder [9] over tokens that exist: the encoder attends only to tokens that carry at least one observed or hidden entry, padded slots are masked out of attention, and within an attended token the hidden and never-observed entries are the learned markers of §2.2 — so the masking is at the level of entries inside a location token, not of whole tokens as in an image MAE. It is a transformer over the example’s tokens — about 900 in the gate phase and 1,200 after hardening (six lags $\times$ column, ellipse and ring $\times$ four groups, plus waist and profiles) — with Perceiver-style cross-attention into a fixed latent set [10], writing a deterministic z of width 128–256. The bottleneck is the token count, not the width. The decoder reads z alone: a query (offset, channel, sphere, depth, lead, footprint) cross-attends to z and emits a mean and log-variance. A decoder that could read the encoder’s latents would let z carry nothing; that path is closed, and its weight is zero rather than small. At the larger rungs the feed-forward block is a mixture of experts routed by sphere and channel group [19].

Objective. A heteroscedastic Gaussian per channel group in its transformed space, decoupled:

$$\mathcal{L} = \|\mu - y\|^2 + \frac{1}{2}e^{-s}(y - \text{sg}[\mu])^2 + \frac{1}{2}s, \quad s = \log \sigma^2, \quad (4)$$

so the mean trains by squared error and the variance by the likelihood with the mean detached, which removes the variance head’s habit of explaining a hard target away. This is the faithful heteroscedastic regression of [12] (the mean’s likelihood gradient replaced by its squared-error gradient, the variance keeping the likelihood’s), with the variance gradient also stopped from reaching the trunk the two heads share; it is not the β -NLL of [11], which re-weights the likelihood by a detached power of the variance and never detaches the mean. $s \in [-8, 8]$; the first 2,000 steps run at $\sigma = 1$; Gaussian in log space for log-normal channels, a hurdle (wet/dry $\times$ log-Gaussian) for rain, logit-Gaussian for fractions. Coverage of the $\pm 1\sigma$, $\pm 2\sigma$ bands is the calibration check.

Weighting across targets. A plain sum over targets whose residual errors differ by orders of magnitude lets the hardest dominate. Standardising weights by marginal variance conflates *varies a lot* with *hard to predict*. The scale used is a reference error — the held-out error of a reference model on that class, which is not the irreducible error but a fixed, measurable stand-in for it:

$$\mathcal{L}_\mu = \sum_k w_k \text{MSE}_k, \quad w_k = 1/\text{MSE}_k^{\text{ref}}, \quad w_k \text{ fixed}, \quad (5)$$

over classes $k = (\text{channel, family, lead})$. With fixed weights this is a Gaussian likelihood with one fixed variance per class; it equals the gradient of $\sum_k \log \text{MSE}_k$ only where the reference equals the current error, and the objective actually minimised is the weighted sum as written. The weights are fitted once, on a calibration fold of training years (never a held-out year and never the run’s own final training error), frozen for the run, and capped at $100\times$ the median weight so that no class dominates through a small reference; at the first run of a tier the reference is the predict-the-mean error, and at later runs the previous run’s held-out error. A run is repeated once if a re-fit after it moves any weight by more than $2\times$. Heavy-tailed classes (rain, chlorophyll) use a robust scale or a Huber error. Logged per class: $w_k \text{MSE}_k$ (near one at convergence) and $1 - \text{MSE}_k/\text{var}_k$. An explicit multiplier for a class a use cares about sits on top, kept separate.

Stability. Collapse is invisible to reconstruction loss, so the effective rank of z on a fixed held-out batch and a scale-invariant probe correlation are logged each evaluation; two sub-threshold readings abort. No quantisation from random initialisation. Gradient clip 1.0; AdamW, $\beta_2 = 0.95$, weight decay 0.1, 2,000-step warm-up, cosine to 10%; bf16 with fp32 master weights; stop at the held-out minimum, at most four epochs [17].

2.6 Stage 2: forecasting embeddings

Inputs. Tokens are the codec’s embeddings: the deterministic z , the count of observations that entered it, offset, statics and season at each sampled cell and lag. The past cone reads eight slices — lag 0, whose embedding already summarises the raw record over lags 0–6, and lags 7, 13, 21, 34, 55, 89, 143 pentads, log-spaced from the first pentad outside the codec’s window to the edge of the two-year window — each with the anchor column, a 24-point footprint whose reach grows at 130 km per pentad ($\approx 1,000$ km at lag 7, the 4,444 km cap from lag 33) and a 24-point ring to the antipode: $8 \times 49 \approx 400$ tokens against twenty thousand for a dense window, which is the ablation. An embedding token has no channel axis, so stage 2’s geometry is per sphere rather than per channel: one drift, waist, growth and memory per 1° cell and sphere, on the same maps, under the same fixed-target guard; the per-channel aperture is a stage-1 device.

Linear backbone. For the anchor column,

$$\hat{z}_{t+\ell} = \mathbf{G}_\ell z_t + f_\theta(\text{cone}, \ell), \quad \mathbf{G}_\ell = \exp(\ell \mathbf{L}), \quad \mathbf{L} = \frac{1}{\tau_0} \log(\mathbf{C}(\tau_0) \mathbf{C}(0)^{-1}), \quad (6)$$

the propagator of a linear inverse model [20, 21] fitted by regularised least squares on training-year embeddings at lag τ_0 of one pentad, retained unchanged, plus a transformer f_θ with a zero-initialised output projection. At step zero the forecaster *is* the embedding-space LIM, without projection or approximation, and it must earn its distance from it held out; fine-tuning $\mathbf{L}$ is a separate arm. $\mathbf{L}$ is a general real matrix: its eigenvalues come in complex pairs, so the backbone carries damped oscillatory modes, and it is non-normal, so finite-lead amplification $\|\mathbf{G}_\ell\|_2 > 1$ is possible while every mode decays — the optimal growth of [20]. The backbone is therefore asymptotically stable (spectral radius of $\mathbf{G}_1$ below one, checked at fit time) and is not called contractive; a parameterisation restricted to a real decay spectrum $\mathbf{U} \text{diag}(e^{-\ell/\tau_j}) \mathbf{U}^{-1}$ would drop the oscillations and, with non-orthogonal $\mathbf{U}$, would still not bound the Euclidean norm. Stability of $\mathbf{G}$ says nothing about the residual model or its multi-year roll, which are scored on their own. Reported before the residual is trained: the backbone’s decoded skill per lead, its finite-lead amplification $\max_\ell \|\mathbf{G}_\ell\|_2$, and after training whether the residual improved it. Damped persistence is the diagonal case and its own null.

Direct multi-lead targets. The mirrored future cone at leads 1, 2, 3, 5, 8, 13, 21, 34, 55, 73 pentads, each a query to the same head, predicted directly, not by chaining: no compounding, amplitude set by the data at that lead, targets where the correlation still changes. Anchor column weight 1, footprint 0.25; the footprint targets make the prediction a coherent field. Beyond a year the predicted anchor embeddings feed back as the recent past: the multi-year roll is a mode, not the objective.

Objective. The decoupled Gaussian on each target embedding, weighted per class by the rule of §2.5 starting from the predict-the-mean bar; plus a decoded-space squared error through the frozen codec decoder at the leads the protocol scores (§5.1), because the embedding metric depends on the codec’s coordinates, the decoder is nonlinear so the decoded mean latent is not the mean decoded forecast, and the scored metric is the decoded one; plus, for the multi-year mode, a two-step pushforward with a stop-gradient on the first step. Two mismatches between training and use are measured on the complete pipeline rather than assumed away: the size of the latent-versus-decoded gap per lead, and the shift a read-out trained on true embeddings meets when fed forecast embeddings (every read-out is scored both ways). If either dominates the error, a larger decoded loss at the scored leads is preferred to a larger model. Under a squared-error mean the amplitude shrinks with correlation, so calibration is a property of the held-out minimum; Murphy’s decomposition [13] $\text{MSSS}_{\text{clim}} = 1 - (1 + a^2 - 2a \text{ACC} + b^2)$, with a the amplitude ratio and b the mean bias in units of the verifying standard deviation, is logged per lead as the check — ACC, a , b and MSSS computed under one set of area weights, the reference being the training-year climatology so the identity is approximate to the extent that reference differs from the verifying sample’s — and a scaling fitted on training-year starts is a diagnostic only.

Regularisation. The globe supplies every regime; the log-spaced cone cuts parameters per example seventyfold; the free part is a residual on a backbone the data pin down. Then: Gaussian noise on input embeddings at 0.7 of the persistence scale; dropout of past points; weight decay; early stopping.

Uncertainty. Three sources are kept apart and never counted twice. The decoder’s variance head gives a per-query predictive variance in the channel’s transformed space. A seed ensemble of three to five forecasters measures seed sensitivity — fitting variability, not the conditional variability of the Earth system — and is reported under that name. The cheap probabilistic baseline is residual calibration: the distribution of decoded residuals by lead and region, fitted on a calibration split of training years and applied to the deterministic forecast, evaluated by coverage and by proper scores in physical units (or under the channel’s stated transform, applied consistently; for a transformed channel the transformed mean, the physical median and the physical mean are three different numbers and each is named). A forecast distribution that beats that baseline has earned its machinery. Once the deterministic level exists, a shared low-dimensional noise vector conditions the head so samples cohere across cells and leads, scored by the fair CRPS of an M -member ensemble [14, 15, 16], $\frac{1}{M} \sum_i |x_i - y| - \frac{1}{2M(M-1)} \sum_{i \neq j} |x_i - x_j|$, which is unbiased in M and so comparable across unequal ensembles. Good marginal scores do not identify a joint distribution: an ensemble trained on marginals reproduced joint spatial structure for one weather architecture [26], which is a reason to try the approach and no guarantee for this one. The joint structure is therefore scored directly — the CRPS of an area average over 10° blocks and of a temporal accumulation or event duration, beside the marginal CRPS — and the diagnostic that shows the aggregate scores can see the problem is to permute ensemble members independently across cells, which preserves every marginal and destroys the dependence. Categorical marginals over a quantised embedding are the alternative where the conditional is two-humped, as an El Niño index at 6–24 months may be.

2.7 Decoding queries

A prediction is a query in the token’s own coordinates: offset in space, depth, time; channel; sphere; source; footprint. A 0.25° cell over a pentad, a 300 m pixel on a day, a station at an hour, a float at its depths, a canopy’s height at a 25 m footprint and a 30 m pixel’s loss year are one decoder, because it was trained on targets with those coordinates and footprints, each scored against the observation that has that support. The domain is what the training targets contained, at the footprints they came with, and the model’s behaviour outside that support is unconstrained — a variance head never trained on a region has no reason to be large there. So support is a flag, not an inference: a query for a channel where the source never covered (SST over land) is *invalid* and is refused; a query for a valid channel at a footprint, depth or place outside the trained range carries an *extrapolation* flag with the answer; and the flags are computed from the training-target registry, not from the model. A functional of many cells is decoded in the way its definition allows. A quantity with a known definition — Niño-3.4, an area-mean SST, a heat content over a stated volume — is the exact area-weighted integral of the decoded field, never a learned read-out. A quantity that is an instrument’s own aggregate with no closed form in the model’s fields — the transport an array measures across its section — is an attention head over the predicted embeddings of its cells, trained on training-year true embeddings, scored on forecast embeddings as well, and never on a spatial mean (geostrophic transport is the contrast across a section, and a mean cancels it). Read-outs are the cheap end of the system and choose between no models. A point store’s own record verifies its forecast; where a licence forbids redistribution, the score is published and the values are not.

3 Data

No source is resampled to a common analysis grid. Each source is stored at its own resolution and cadence in one of three tiers; where a source is aggregated at build time — the pentad mean of a daily field, the block mean of 4 km colour under a 0.25° point — the aggregation is stated with the group and is the footprint its token carries; the sampler turns all of them into one token at read time; the decoder is queried at a location and a footprint.

3.1 The 0.25° pentad grid

Grid and time. A point-aligned 0.25° grid, $721 \times 1,440$ cells, rows south first, longitude wrapping; a five-day bin (pentad) from 1982-01-01, bin b covering days $[5b, 5b + 5)$; 3,142 bins, the last opening 2024-12-31 and holding one day of source data rather than five. A bin is an averaging window, not a date. A coarse 1° grid (181×360) is defined the same way; a coarse channel is read at its nearest coarse point, so sixteen fine cells see one number. That is the model’s honest view and is not interpolated.

Groups. Each group is an array $[T, H, W, C]$ in float16, bin-major, at its native grid and cadence, so one bin of one group is one contiguous byte range of one file (14.5 MB for the ocean surface, 2.0 MB for the atmosphere) and a reader fetches a pentad with a single HTTP range request. Four groups carry 56 channels:

- **ocean surface** (g025, seven channels), 0.25°, every pentad: current speed and its two components, sea-surface height and log mixed-layer depth (GLORYS, 1993–), sea-surface temperature and sea-ice concentration (OISST, 1982–);
- **atmosphere and land** (g100, fifteen), 1°, every pentad, every surface: wind stress and its within-pentad standard deviation (four), 2 m air temperature, 10 m wind (two), surface pressure, log precipitation rate, log snow water equivalent, soil moisture and temperature (0–10 cm), latent and sensible heat flux, skin temperature (NCEP; ERA5 is the intended replacement);
- **ocean interior** (rg100, thirty-two), 1°, monthly: temperature and salinity at sixteen pressure levels, 10–1,900 dbar (Roemmich–Gilson Argo, 2004–), each month stored in the pentad holding its fifteenth, 252 rows. The field is an objective analysis of every profile in the calendar month and is released after the month ends, so its support is the whole month and its availability is the month’s end plus about a week; the mid-month bin is where it is stored, not when it was known (§3.5);
- **ocean colour** (oc025, two), 0.25°, 1,997 rows on an offset axis from bin 1145 (1997-09-04): block mean of $\log_{10}$ chlorophyll over the 6×6 OC-CCI (the European Space Agency’s Ocean Colour Climate Change Initiative) 4 km cells under a point, per day and then over the pentad’s days, and the observed fraction of cell-days — the evidence behind the value, missing exactly where the value is, never zero; 855,742,985 finite values per channel, 41.3 % of the group, colour over estuaries and lagoons the land mask calls land kept and counted;
- **statics:** surface class (702,642 ocean, 226,495 land, 107,074 ice sheet, 2,029 inland water; ice over ocean on a shelf) and cell-mean elevation from ETOPO 2022, both complete in all 1,038,240 cells.

No channel is masked by sphere: a land cell reads its skin and air temperature and soil, an ocean cell its surface fields, every cell the shared atmosphere. A channel is a quantity and an instrument, never a sphere; observed sea-surface and land-surface temperatures are separate channels.

Standardisation, transforms, missing values. Stored values are $x' = (x - \mu)/\sigma$ per channel. At load time every channel becomes a standardised anomaly against a day-of-year harmonic climatology (annual and semi-annual) fitted on training years only, over a training-year standard deviation; no held-out year touches any statistic. Log-normal quantities (mixed-layer depth, chlorophyll, precipitation, snow) are logged first; bounded fractions (sea ice, soil moisture) go through a logit. A value that does not exist — a land cell’s SST, a month without Argo, a cell under cloud — is not zero: every token carries an observed flag per channel, and an unobserved entry is masked from loss and input. Holes in the grid are holes in the sources, counted per channel and bin at build time and again by streaming every published file: 1.554×10^{10} observed values; every channel full over its own record, except where a source is empty — the reanalysis ocean before bin 804 (1993-01-03), sea ice in bins 433–439 (a change of microwave sensor), 2846 and 3090, colour in bin 1153 — and the one-day last bin, empty in the two groups that require three days of a pentad.

3.2 Finer sources in three storage tiers

Three facts decide how finer sources enter. A coarse value copied into fine cells is a lie: nine copies are not nine measurements. A fine value averaged into a coarse cell discards what the source was bought

for: 4 km colour into 0.25° keeps one part in thirty-six. And missing is not damage, nor is far away: the nearest profile’s value depends on its distance, which a grid cell hides. Sources are therefore stored by density, in three tiers under one registry:

- **G, gridded dense** — fields at $\lesssim 5$ km, daily to monthly, near-complete: colour at 4 km, SST at 5.6 km per sensor and 2.2 km merged, sea ice at 6.25 km, 4 km cloud-top temperature, 0.1° daily rain, the daily 0.05° land temperature, snow and reflectance, 500 m leaf area every eight days, and the annual maps — forest loss, canopy height and biomass, the leaf calendar — at 30–500 m. A group finer than 0.25° is sharded: one file per group and pentad holding F frames (one per day, or three hours, or one per year), each frame cut into 256×256 -pixel tiles of the native grid, each tile compressed on its own with `zstandard`; beside the shard an index of (byte offset, length) per frame and tile, where an empty tile has length zero and is never stored and a frame absent from the source has offset -1 , so a missing day is recorded, not silently skipped. Tiles are written frame by frame without gaps, so a frame is two range reads — its slab of the index, then one contiguous span of the shard — and a cone decodes only the tiles it touches (578 tiles a frame for the 4 km colour grid, 11.6 % of its pixel-days valid). Footprint is a constant per group.
- **P, points, profiles, tracks** — Argo and BGC-Argo, every non-Argo profile since 1772, ship and buoy reports since 1662, bottle chemistry, drifters, tropical and open-ocean moorings, the overturning arrays’ instruments, surface CO_2 , along-track altimetry at 7 km and SWOT swath cells at 2 km, column CO_2 soundings, wave height, scatterometer wind cells; on land, stations, balloon soundings, tide gauges, buoys, gliders, flux towers, fire detections, rivers, soil probes, laser footprints of the canopy, fluorescence soundings, species occurrences. A store is a set of columns — time in integer seconds, latitude, longitude, values in raw units, mask, quality, hashed platform, per-row footprint — sorted by (bin, time) with a compressed-sparse-row index of row offsets per bin, so the k nearest observations in space-time are one bounded scan that stops at the anchor’s own bin and never reads the future. The bin is derived in integers, $b = \lfloor t_s / 432,000 \rfloor$ with t_s the seconds since 1982-01-01T00:00 UTC, floored toward minus infinity, so pre-1982 rows keep negative bins (drifters from 1979, carbon from 1957, ship reports from 1662) and the loader, not the store, clips to a tensor’s axis; each store’s index runs over its own first bin, which is not zero. The seconds are `int32` (1913 to 2050), or `int64` in a store whose record begins before 1914 (schema 3 against schema 2). They replaced `float32` days, which resolve 21 s in 1993 and 84 s by 2015 — up to 316 consecutive one-second altimeter samples shared one timestamp, 550 km of track inside which Δt ordered nothing, and 908 carbon rows within 84 s of a pentad boundary were filed one bin late; integer seconds cost the same four bytes.
- **T, tiles** — imagery at ≤ 1 km (Sentinel-1/2, Landsat, NISAR, OLCI, SLSTR, VIIRS, MODIS, the L- and P-band radar mosaics) and the 10 m and 1 m canopy-height maps are not stored as values. The store is a scene-date catalogue (footprint, sensor, time, cloud fraction, location) and a token cache: a local codec turns one tile-date into a few 64-dimensional vectors, stored as a tier-P store whose values are those vectors, footprint the tile and the day. AlphaEarth’s published embedding [2] — 64 numbers per 10 m of land per year — enters by the same path as a ready-made cache, so the programme’s own tile codec can be measured against it with everything else fixed.

One time axis (the pentad bin; tier-P rows keep their exact time, so Δt is fractional); one registry per family (tier, layout, cadence, channels, footprint constants, first bin, checksums, sources, licence, builder commit — a consumer dispatches on the tier), in which every group also carries its *source contract*: `support_start` and `support_end` (the interval of measurements a value summarises, relative to its stamp — zero width for an instantaneous reading, the calendar month for a monthly analysis, the fit period for an annual map), `available_at` (the earliest time the value could have been used, from the producer’s release latency; equal to the stamp for a real-time source), `product_version`, and `lineage` (the measurements a derived product was fitted to or assimilated, by store name — the Roemmich–Gilson field to the Argo profiles, the reanalysis to the profiles and the altimeter, a canopy map to the laser footprints, a tile cache to the imagery and to whatever supervised it). Annual maps, retrospective analyses, climatological backgrounds and pre-trained embedding caches carry these fields like any other source, since an anomaly climatology recomputed on training years does not remove future information already inside an upstream product; one sampler contract (`gather(anchor, family)` returns tokens per group under that group’s cone family, with a fixed slot count so batches stay rectangular). Licences decide the track. Each group’s licence block records `redistribution` (yes, with attribution, or no) and

store	what it is	producer	space	time	record	tier
seaice_asi	daily sea-ice concentration, microwave	Univ. Bremen, AMSR-E/2	6.25 km polar	daily	2002-11, 2012-	G
oc4k	daily chlorophyll, attenuation, reflectance	ESA OC-CCI v6.0	4.6 km	daily	1997-09-	G
sst_cci05	daily SST per sensor	ESA SST CCI v3.0	5.6 km	daily, day/night	1979-07-	G
sst_acspo02	daily SST merged across sensors	NOAA ACSPO	2.2 km	daily, AM/PM	2000-02-	G
irtb	geostationary cloud-top temperature	NOAA CPC	4 km	30 min (3-hourly first)	1998-01-	G
precip01	daily satellite rain (IMERG)	NASA/JAXA GPM	11 km	daily	2000-	G
wind12	scatterometer ocean wind	OSI SAF; QuikSCAT	12.5 km swath cells	per pass	1999-2009, 2007-	P
swot	sea-surface height, two 50 km swaths	NASA/CNES SWOT	2 km	per pass, 20.86 d	2023-07-	P
swh	wave height along altimeter tracks	ESA CCI Sea State	≈7 km	1 Hz	1991-08-2023-12	P
icoads	ship and buoy weather reports	NCEI / NCAR, ICOADS 3.0	point	per report	1662-	P
wod	T, S, O ₂ profiles, Argo excluded	NCEI, WOD	profile	per cast	1772-	P
glodap	bottle carbon, nutrients, oxygen	GLODAPv3	bottle	per cruise	1972-2023	P
bgcargo	floats: O ₂ , nitrate, pH, chlorophyll	Argo GDAC	profile	≈10 d per float	2002-	P
oceansites	open-ocean moorings	OceanSITES	mooring	hourly or finer	1980s-	P
deep_arrays	the four overturning arrays' instruments	RAPID, OSNAP, SAMBA, MOVE	mooring	15 min-1 h	2000/04/09/14-	P
xco2	column CO ₂ soundings	OCO-2/3; GOSAT	< 3 km ²	per sounding	2014-06-	P

Table 2: The global fine-granularity family. Each store is kept at its own resolution and cadence and enters through the token of §2.2 with its own footprint. A one-month test fetch through the real builder replaces every size estimate before a store is built. State on 22 September 2026 (§3.4): `glodap`, `wod`, `icoads`, `bgcargo`, `oceansites`, `swh` and `oc4k` (to 2022-12) are built and public; `seaice_asi` is built on the private track (from 2012-07, redistribution not yet granted); `pace4k`'s parts are parked; `sst_acspo02` and `swot` are probed and not built; `irtb` and `xco2` wait on a data-access approval; `sst_cci05`, `precip01`, `wind12` and `deep_arrays` are designed and not in the registry.

`derived_works` (free, non-commercial or restricted) from the licence's own clause, and `distribution` is public by default and private whenever redistribution is not granted: a private store keeps the same arrays and layout in a private repository, is listed in the public registry with its sizes and checks so a reader without access skips it by the field rather than by a failed fetch, and is pulled for a job and never kept on a rented machine. What may leave the private track is a trained model and its scores where `derived_works` allows, never the values, and a result resting on a private store names it. A non-commercial source whose licence permits redistribution stays public and carries the restriction in its fields.

The global fine-granularity family (1.gf, Table 2) is the instance of tiers G and P over the ocean and air at ≤ 10 km and ≤ 5 days; a land-and-coast family (1.0.tf) holds the station networks, gauges, towers, fires, rivers, the 0.05° land fields, the imagery catalogue, the land-change targets and the biosphere stores of Table 3; a third (0.9.tf) is that family with the catalogue replaced by the AlphaEarth cache (§3.4 gives what is built). Admission: raw measurements first, a derived product only where it carries measurements nothing else does or is itself a target a user will ask for (the annual maps), no gap-filled analyses; four named exceptions slightly coarser than the rule (12.5 km scatterometer wind, 11 km rain, one 10.5 km CO₂ sounder, eight-day leaf area); and every store is scored on the read-out it was admitted for, at equal token budget with and without it, with its incremental cost recorded. The verdict has three values, not two: *useful* (it lowers the error beyond the paired interval), *not useful at this budget*, and *inconclusive* (the read-out lacked the power to say). A store that is not useful at this budget or inconclusive is deferred — its registry entry, provenance and cost stay — and is not called uninformative, because an absence of measured benefit on a low-power read-out is not an absence of information. An observation count is not independent information: a long altimeter track or a dense buoy array can dominate a loss or a nearest-neighbour set without adding a regime, so sampling and verification of tier-P stores are platform- and track-aware, and a store's benefit is inspected conditional on the companion channels that make it useful.

The local codec. A 64×64 tile at 10 m is 4,096 pixels of 13 bands and a 0.25° cell holds $\sim 7,000$ of them; they cannot be sampling points. The local codec is a masked autoencoder over one tile-date,

store	what it is	producer	space	time	record	tier
gedi	laser relative-height profile, cover, biomass per footprint	NASA GEDI, ISS	25 m every 60 m	per shot	2019-04–	P
icesat2	laser terrain and canopy height per segment	NASA ICESat-2 ATL08	100 m	per pass	2018-10–	P
sif	chlorophyll fluorescence soundings	TROPOMI; OCO-2/3	7×3.5 km; 1.3×2.2 km	per sounding	2018-05–; 2014-09–	P
flux	tower CO ₂ , water and heat exchange	FLUXNET, 806 towers	tower	half-hourly	1991–	P
fire	active-fire detections	FIRMS, MODIS/VIIRS	375 m–1 km	per overpass	2000-11–	P
gbif	species occurrences, taxon as platform	GBIF snapshot	per-record uncertainty	day or month	1600s–	P
lai500	leaf area, absorbed light	MODIS/VIIRS	500 m	8-day	2000-02–	G
pheno500	green-up to dormancy dates, amplitude	MODIS MCD12Q2	500 m	annual	2001–2025	G
canopy30	canopy height 2020	GLAD, Landsat fitted to GEDI	30 m	one year	2020	G
biomass100	above-ground biomass and its error	ESA CCI v7; GEDI L4B	100 m; 1 km	annual	2005–2024	G
lossyear	the year each pixel lost its forest; tree cover 2000	Hansen GFC; JRC TMF	30 m	annual	1990/2001–	G
canopy_ref	10 m and 1 m canopy height, catalogued	ETH; Meta/WRI	10 m; 1 m	one year; a decade's composite	2020; 2009–20	T
cat_palsar	L- and P-band radar, catalogued	JAXA PALSAR-1/2; ESA BIOMASS	25 m; 50–200 m	annual; per pass	2007–; 2025–	T
pace4k	hyperspectral colour and phytoplankton groups	NASA PACE OCI	4 km	daily	2024-03–	G

Table 3: The biosphere stores of the land-and-coast family (**pace4k** belongs to the ocean family). Same conventions as Table 2; the annual maps are targets, the rest inputs. State on 22 September 2026: **fire**, **flux** (26 of 806 towers unread) and **lossyear** are built and public; **gedi**, **icesat2**, **lai500**, **pheno500**, **canopy30** and **pace4k** have parts parked and no assembled store; **sif**, **gbif**, **canopy_ref** and **cat_palsar** are registered and not built (**gbif**'s non-commercial part is registered on the private track, unbuilt); **biomass100** is designed and not in the registry.

conditioned on sensor metadata (resolution, incidence or sun angle) with a learned source embedding so one codec serves optical and radar. A same-place-different-date contrastive term would make the token invariant to exactly the changes a forecast must retain, so it is not switched on by default: the token's sensitivity to real change (a clearing, a burn, a green-up) is measured first, on the loss-year and phenology targets, and any invariance term is added only with augmentations that preserve the state of interest. It emits $k = 4\text{--}8$ tokens of 64 dimensions, cached once, reused by every anchor whose cone holds the tile; it sees one date, and the cone codec supplies time. Tile size is a knob: 640 m tiles at seventy clear dates a year are ~ 20 TB of tokens a year globally, 2.56 km tiles sixteen times less. Imagery is streamed once to train and once to encode, never kept.

3.3 Biosphere observations

The living surface is observed by few instruments and fitted by many maps. The measurements are the laser footprints (GEDI [40] to $\pm 51.6^\circ$, ICESat-2 [41] beyond), which alone see a canopy in three dimensions; fluorescence soundings [42], emitted by photosynthesis itself; the towers' eddy-covariance carbon flux [43], the long in-situ record; leaf area every eight days; fire detections; and 3.7 billion dated, located species occurrences [44] — sparse and biased, and the only global record of where life is. The annual maps — forest loss [45], canopy height [46, 47, 48], biomass [49], the leaf calendar — are each fitted to one of those measurements and are admitted as targets: what a user asks for, at the map's own footprint and year. The first model assumes the dominant predictors of a biosphere channel are the local atmosphere and soil, which every token already carries, and gives land no drift; that is an assumption about the first model, not about land — seed dispersal, river transport, fire spread, hydrology, disturbance and management all move signals across land, and fire already has the wind as its carrier. Species occurrences enter as a channel only once a model of observation effort exists: a record says a species was seen, its absence says nothing about whether it was there, and a generic Gaussian channel cannot tell absence from lack of reporting; until then **gbif** is stored and not read.

3.4 The families as built

The design is built as numbered families, each a set of stores under one registry of the same shape; this is their state on 22 September 2026. Family 7.2 is the grid of §3.1 (`family7_global025_pentad_12`): the groups `g025`, `g100`, `rg100` and `oc025`, 3,142 pentads from 1982-01-01 to 2024-12-31, 61.16 GB. Family 8 is the Argo profiles themselves, 2,678,439 from 2004 to 2024 with $C = 32$ (temperature and salinity at the sixteen Roemmich–Gilson pressures). Family 10.2 is five tier-P stores with the integer-second time column: `gdp`, 48,480,798 six-hourly fixes of the Global Drifter Program; `gtmba`, 1,001,282 daily rows of the tropical moored buoy arrays; `socat`, 41,830,675 underway measurements of surface-water CO₂ fugacity from the Surface Ocean CO₂ Atlas; `slatrack`, 2,030,800,176 one-second along-track altimeter samples of sea-level anomaly from 29 mission datasets; and `fishing`, 617,164,038 vessel-cell-days of apparent fishing effort that Global Fishing Watch derives from the Automatic Identification System, the collision-avoidance radio of large vessels — under a Creative Commons Attribution-NonCommercial licence (CC BY-NC 4.0), its fishing-hours channel the one value in the family that is a classifier’s output rather than a measurement. With Argo that is 2,741,955,408 tier-P observations. The registry `tensors/family10_2/family10.json` lists these ten groups — the grid’s four referenced by its manifest’s hashes, never copied, and six point stores — with an empty list of groups it could not read, and a `siblings` block naming the three fine families, so a consumer that dispatches on the tier discovers them from the one entry point. Family 1.gf is Table 2: 13 stores, 7 built and public, 1.1 billion ship and buoy reports (`icoads`) and 1.7 billion wave-height samples (`swh`) the largest. Family 1.0.tf is the land-and-coast family: 33 stores, 17 built and public — stations, balloon soundings, tide gauges, buoys, gliders, fires, towers, 0.05° pentad rain, 30 m forest loss and eight scene catalogues. Family 0.9.tf inherits every 1.0.tf store but the catalogues by reference and adds four stores of the AlphaEarth embedding, none of them started. Everything public is in the Hugging Face dataset `chfrank/earth-tensors`; the private track, `chfrank/earth-tensors-private`, holds the built `seaice_asi` and `tide_private`.

3.5 The evaluation contract: periods, availability, lineage

Periods. The record is divided into three kinds of period, and every learned component — codecs, tile caches, the linear inverse model, read-outs, loss weights, calibration — obeys the same division. *Training* is 1982–2020 less the *development* years 2009 and 2017, which are held out of training, climatology and every statistic and are where architecture, sources and training choices are compared. *Retrospective evaluation* is 2021–2024: no model is trained on it and no statistic is fitted on it, but these years have informed design decisions of this programme — probes and reconstructions were scored on them, and 2023 served as a development year in the pilot — so a score on them is labelled retrospective and is evidence about skill conditional on the reconstructed state, never a first look. *Prospective test* is 2025-01-01 onward: no statistic, model or decision has touched it, the tensors do not yet contain it, and it is scored as issued — forecasts for it are produced and archived with their issue time before the verifying observations are ingested, and each is scored once. The prospective record grows by one year each year; nothing is promised about a pristine period that has already been seen.

Two evaluations, kept apart. A *retrospective state forecast* may read every declared retrospective analysis product at its stored bin, with the product’s support audited, and establishes skill conditional on that reconstructed state. A *forecast as issued at t* reads only rows with `available_at` $\leq t$ and establishes deployable skill. Every score names which of the two it is. The difference is not small: the monthly Argo analysis (§3.1) is stored mid-month, summarises the whole month and is released a week after it, so a retrospective forecast issued mid-month reads a month of the future’s profiles that an as-issued forecast does not; the reanalysis ocean assimilates observations from after the bin it is stamped with; an annual map summarises a year. The size of each such gap on the scored metrics is measured on the complete pipeline, not assumed.

Lineage holdouts. Holding out a measurement does not hold out the products fitted to it: a hidden Argo profile still shapes the gridded Argo analysis and the assimilating reanalysis of the same month, a hidden laser footprint may have supervised a canopy map or a tile cache. For a forecast of later independent observations this is not a leak; for a claim of independent reconstruction or of generalisation across sources it is. Such a claim is therefore scored under a lineage holdout — the `lineage` field groups a measurement with every product derived from it and the group is withheld together — and is not

rung	values	tokens	ceiling	what it takes
one basin, 0.25° pentads, 42 channels	2.5 B	0.04 B	0.12 B	the pilot’s tensor
global pentads, four groups	13.8 B	0.21 B	0.7 B	built (§3.1)
+ the shared channels of every sphere	91 B	1.4 B	4.5 B	ERA5, radiometers, scatterometers, gravimetry, soil moisture
daily cadence, all spheres	455 B	7.0 B	23 B	a 1.3 TB tensor and a streaming loader
kilometre tier	3.5×10^5 B	5.4×10^3 B	—	the token cache; compute-bound

Table 4: The data ladder in observed values; *tokens* counts each location–pentad once (at 65 channels each) and bounds the distinct information, not the compute (see text); the *ceiling* is one parameter per twenty values [18], a planning heuristic borrowed from language models and an upper bound because values are correlated, not a scaling law for this system.

made otherwise.

Admissibility and its check. An example is admissible only if every bin its forward pass can see — every past token, every future target, every point observation’s exact time — lies in the training period. The check is deterministic, not sampled: for each anchor bin the earliest input lag and the latest target lead are compared with the period boundaries, and the representative boundary cases (an anchor whose two-year window crosses a development year, whose target year crosses the retrospective boundary, whose nearest profile falls in a later bin) are enumerated and refused before a run. A random draw of anchors is a diagnostic that runs beside it, never the certificate.

Longitude holes. Meridional blocks are withheld from targets and from every fitted statistic; inputs inside them are not hidden, and the underlying observations of derived products are not touched. What a longitude hole measures is therefore the forecast head’s generalisation to places whose targets it never fitted while it could still read their inputs — one kind of generalisation, named as such and scored apart from interpolation — not source generalisation and not independent reconstruction, which are the lineage holdout’s.

The manifest. One machine-readable split manifest per run records the period boundaries, the source–contract fields of every group read, the lineage groups withheld, the evaluation mode (retrospective or as-issued) of every reported score, and the outcome of the admissibility check; a result without its manifest is not reported.

4 Training and scaling

Data is priced in observed values and their independence, not bytes. A 0.25° cell is correlated over hundreds of kilometres and, at the surface, months, so a basin’s 10^9 values are 10^6 – 10^7 independent ones — the regime in which a small model equals a large one. The globe adds regimes, the spheres add physics, daily cadence is mostly the same sample five times, the kilometre tier adds new variance. Order: globe and spheres, then footprint, cadence last (Table 4).

Each rung is bought on the previous rung’s read-out under the frozen protocol with the null ladder beside it. The test that a rung is over-parameterised is not a step count — a held-out minimum “inside 2,000 steps” depends on batch size and learning-rate schedule and does not travel between runs — but a learning curve: the held-out loss against examples processed, and, before any move from millions to billions of parameters, a 2×2 of two data extents at fixed size and two sizes at fixed data and compute, with examples and tokens processed recorded per cell. A rung whose larger cell does not beat its smaller one at equal data, and whose larger-data cell does beat its smaller one at equal size, is data-limited and the next data rung is bought; the reverse buys parameters; neither pattern points at the representation, the objective, the optimisation or the target noise rather than at scale, and those are tested before either is bought. L1: a 7–125 M codec on the 0.25° pentad grid, fixed against learned geometry. L2: 0.3–1 B codec and forecaster with the tier-P stores; the width ladder is re-measured here, the one test that separates data-limited from compute-limited. L3: 1–4.5 B, sphere-routed experts, all

spheres: do land and atmosphere improve the ocean’s rolled skill and vice versa. L4: daily cadence, the first compute-bound rung. L5: the kilometre tier, after L4 is positive.

Cost. The forward cost of the codec is not $6N$ per tensor token, because a cone codec reads each cell-pentad once per anchor whose cone contains it. The unit is the *anchor*: an example of ≈ 900 input tokens cross-attended into a latent set of 64–256 vectors. The measured anchor is a 7 M-parameter cone codec of 748 tokens per anchor at 20,000 steps $\times$ 256 anchors, 58 minutes on one RTX 4090, or 1.1×10^6 input tokens a second; at that rate the input projection and cross-attention, not the latent stack, set the time. Scaling the latent stack to 10^9 parameters at the same anchor budget (5.1×10^6 anchors, 4.6×10^9 input tokens) costs of order 10–130 RTX-4090-hours depending on where the parameters sit — the lower figure if the latent stack alone grows, the upper if $6N$ is spent on every input token — and a sustained 66 TFLOP s^{−1} (40% of the card’s dense bf16 peak) is the throughput assumed. That budget touches about 0.3% of the $\approx 1.8 \times 10^9$ ocean cell-pentads as anchors; a pass that uses every one of them once is 10^4 – 10^5 RTX-4090-hours and is not planned. “Epoch” below therefore means a pass over the anchor budget, and the four-epoch rule [17] — a language-model heuristic for repeated data — is not the binding constraint; the number of distinct observed values the anchor budget has seen is, and it is logged. Every stated cost is replaced by measured anchor throughput, encoder and decoder token counts, loader time and cache-construction time before the rung is bought. The aperture and ring cost one multiply per token. The gather is host-bound and batched by latitude row. Every artefact is written incrementally to durable storage while a run lives. A configuration is a named recipe; a resumed run derives its architecture from its checkpoint. One seed is a direction, never a level; the first result at a new tier buys its replication, and a probe-scored claim needs three seeds.

5 Evaluation methodology

5.1 Protocol and baselines

Periods, evaluation modes and lineage as in §3.5; every score is labelled retrospective or as-issued. **Starts.** Forecasts are issued from every pentad of an evaluation year where inference cost permits, and from no fewer than one start a month; three starts a year under-sample the seasons and the dependence on the forecast origin. Overlapping forecasts are not independent: intervals come from a block bootstrap over starts with blocks that preserve the temporal dependence of the lead being scored [22], reported with their sensitivity to block length and with the number of distinct years and events behind them, and a paired comparison is always paired by start and seed. A few years support no claim about multi-year timescales however many pixels they contain, and training seeds add optimisation variability, not climate histories. **Scores.** Per lead and channel, under one explicit set of physical area weights and fixed channel and region weights: mean-squared skill against climatology, raw and damped persistence; anomaly correlation; amplitude ratio and mean bias; per scope (trained and held-out longitudes). Beside the dense field, a fixed *decoded panel* that the dense score cannot stand in for, because more pixels are not more independent El Niño events or overturning transitions: area-mean SST over named regions (Niño-3.4, the North Atlantic subpolar gyre, the tropical Indian Ocean), upper-ocean heat content to 700 dbar where Argo coverage supports it, the Niño-3.4 index as an exact area-weighted integral, each at leads of one, three, six and twelve months; and for one supported event class (an El Niño onset, defined by a threshold on the index) the probability of exceeding the threshold, the event’s duration and its accumulated anomaly. The panel is fixed here and not extended before the first result. **Slow background.** For seasonal and interannual leads every score is reported twice: against a training-only evolving climatology (a linear trend plus the harmonic seasonal cycle, fitted through the issue time) and against the stationary climatology, so trend reproduction and internal-variability prediction are visible separately; the trend is part of the target and is not removed. **Proxies.** A score is named for what it measures: a surface-CO₂ error is not an air–sea flux skill, a velocity or a transport at one section is not the overturning, a canopy height at a footprint is not a vegetation change; each is scored as the precursor or proxy it is unless the named quantity is scored with its own observation and aggregation assumptions stated. **The null ladder** beside every read-out: climatology, persistence, damped persistence (per-cell AR(1) on training years), the LIM in data and in embedding space [20, 21], a direct multi-lead regression on compact features, an analogue forecast, the previous rung. **Decision rule.** Each comparison names one primary score — the area-weighted mean over the scored leads and channels of the mean-squared skill against damped persistence, unless stated otherwise — with a practical improvement threshold

and an equivalence margin fixed in advance, and a short list of non-degradation checks on the panel. A difference inside the equivalence margin at the paired interval is “equivalent”; a difference the interval cannot resolve is “inconclusive”, which is not evidence of equality; an inconclusive comparison keeps the simpler arm operationally and reports the uncertainty. Requiring a win at every lead and every seed is not a rule here: it is brittle and biased to the shortest leads. **Evaluation inputs.** The query and target set of an evaluation is frozen and identical across arms; each trained model reads it through its own intended, admissible input sampler, so a model with a learned cone is evaluated with the cone it learned. Forcing every arm through the fixed sampler tests behaviour under an input change, not the intended deployment, and is reported as a separate diagnostic where it is run. **External baseline.** On the atmospheric channels at leads up to fifteen days the external baseline is FGN [26], a data-driven ensemble trained on ERA5 and fine-tuned on the ECMWF operational analysis, whose released checkpoints (WeatherNext 2) are run here over 2023 from the analysis they were fine-tuned on; its 6-hourly 0.25° members are averaged to the pentad bins and the 1° cells the atmospheric group holds, so both models are scored on the same output support. Equal output support is not equal input information: FGN starts from an instantaneous global analysis, this model from a pentad observation stream, so the comparison is a *system benchmark*, not a matched comparison, and it is labelled one. Its conventions are fixed: the issue time is the end of the input averaging window; each forecast’s valid interval is the pentad it means; precipitation is the pentad accumulation of each system’s own rate, averaged the same way. Scores: fair CRPS of the ensemble, ensemble-mean RMSE and spread-skill ratio, per channel and lead, and CRPS pooled over 10° blocks. Configurations are compared on the dense field and the panel; the five read-outs are checks with limited power, kept without letting an inconclusive read-out veto a representation. A winner is replicated with a second seed before it is written up; a negative result is reported like a positive one.

5.2 Pre-registered comparisons

(0) **The forecasting benchmark**, first, before any geometry is learned or any store admitted, because a reconstruction that works, a sampling pattern that is physically plausible and billions of measurements do not by themselves show that the representation improves a forecast of the quantities of interest beyond cheap models given the same information, and the pilot found a current-state read-out from the codec at parity with the raw fields and a linear model stronger beyond two weeks [1]. Existing data, the fixed hourglass, the smallest practical model, one common information set — the same forecast origins, variables, history, targets and preprocessing — and four arms: **B1** damped persistence and a data-space LIM on compressed (EOF) features, the skill already available cheaply; **B2** a direct multi-lead regularised regression on compact raw or EOF features over selected lags, which says whether reading history and fitting each lead directly explains any gain before it is credited to nonlinear dynamics or to cone geometry; **B3** the fixed codec plus an embedding-space LIM, which says whether the representation is a better compression for forecasting; **B4** B3 plus a small nonlinear residual, which says whether there is predictable nonlinear signal beyond it. Scored in decoded physical space, on SST at one, three and six months and one subsurface quantity (heat content to 700 dbar), with the analogue baseline where affordable and ENSO and transport as secondary checks; one land task is added only where a record long enough exists. Runtime and paired uncertainty are part of the table. The decision this buys: if B3 does not beat B2 on the primary score, a smaller codec and a modest decoded forecast supervision are tried before any source is imported or any geometry learned; if B4 does not beat B3, the residual is not grown. One error decomposition accompanies the table — by lead, season, region, slow background against anomaly, and on one scientific aggregate — to say whether the limiting factor is the representation, the dynamics, the calibration or the data coverage.

(1) **Hourglass against snapshot codec**, both frozen, same forecaster: one-step ratio, future-cone families against their bars, the T1–T2 gap.

(2) **Geometry**, run only if (0) shows a limitation that geometry can address, at equal token and compute budget, same tensor, codec size and steps, three seeds each, the target set fixed and shared: **A0** fixed (the control); **A1** learned per channel, hierarchical, from A0’s initialisation; **A1g** learned per group, which A1 must beat to earn its extra estimation burden; **A2** as A1 with the ocean-surface drift primed at $-\mathbf{u}_{\text{clim}}\Delta t$. Read-outs, fixed now: **R1** the primary score of §5.1 on T1, with the per-lead losses against their bars as the non-degradation checks; **R2** the T1–T2 gap; **R3**, descriptive only, per-channel drift against its expected carrier over the world ocean where the carrier exceeds 0.2 ms^{-1} — currents against $-\mathbf{u}_{\text{clim}}$, height’s zonal drift positive (source to the east), ice against the reversed wind turned

5–30°, SST against the reversed wind at lags 0–1, deep levels against the reversed reanalysis velocity at their depth; **R4**, descriptive only, at an anchor north of the 26.5°N array, surface drift south-west up the Florida Current and the drift of the $\geq 1,000$ dbar channels north up the deep western boundary current, with attention on deep profiles north and shallow south. Verdict: A1 is adopted if it beats A0 on R1 by more than the practical threshold at the paired interval with no check degraded; A1 within the equivalence margin of A1g keeps the per-group form as the cheaper equal; A2 beating A1 with A1 at A0 sets the geometry from climatology and is read as a statement about optimisation, regularisation and initialisation, not as proof that gradients cannot find upstream; none winning keeps A0. R3 and R4 are reported whatever the verdict and decide nothing. First ablations after: A1 without the ring, and A1 with the ring replaced by a coarse global summary of equal token budget.

(3) Forecaster: the LIM backbone alone (its decoded skill and finite-lead amplification reported first), backbone plus zero-initialised residual, the residual with a fine-tuned **L**, and the transformer alone; direct multi-lead against rolled one-step; log-spaced against dense lags. A constrained propagator (a real decay spectrum, or damped 2×2 rotation blocks) is an arm only if the unconstrained backbone shows transient growth that costs skill, with the trade-off stated: suppressing growth or oscillation may remove predictability.

(4) Footprint and tiers: coarse groups as offset tokens with true footprints against nearest-cell copies; the stores of Table 2 and the point families with against without, per read-out — El Niño by Niño-3.4 at six months against warm-water-volume persistence; carbon by held-out surface CO₂ against a climatological null; overturning by the velocity loss with drifter tokens against without; SST by the level-3 cell loss with drifter and ship tokens against without; the land biosphere by held-out tower carbon flux against a per-site day-of-year climatology and damped persistence, and by canopy height at held-out laser footprints against the 2020 map and the footprint’s previous value — under the lineage holdout of §3.5, since the map and the AlphaEarth cache were fitted to the same laser — with fluorescence at held-out soundings as the dense check — and per store on the read-out it was admitted for; the coastal tile codec with against without tile tokens, three seeds; AlphaEarth’s cache in place of the tile codec, scored on the canopy read-out as well, since that embedding was fitted to the same laser.

(5) Scale: the width ladder at L2; sphere-routed experts against a dense model of equal active parameters at L3.

(6) Weather: the shared atmospheric channels — 2 m temperature, 10 m wind, surface pressure, precipitation — at leads of one, two and three pentads against FGN on 2023 (a retrospective year here and FGN’s own test year), with damped persistence and climatology beside both. The forecaster is not a weather model: it reads pentad means and predicts pentad means, and FGN is scored on the same means. The comparison says how much of the medium-range atmosphere a model built for the coupled system at pentad cadence gives up, and whether the ocean, land and biosphere stores buy any of it back (arms with and without them).

(7) Completion: T4c on its fixed pair list against its predict-the-mean bar, the one comparison that licenses a claim of cross-variable inference.

Build order: the evaluation contract and its manifest (§3.5); the forecasting benchmark (0) with the fixed hourglass and the error decomposition; the forecaster’s backbone and residual (3); then, and only where (0) has located a limitation they address, geometry (2), task mix and aperture; then one source family chosen against a documented error, with its lineage-safe holdout and its incremental cost (4); replication of any promising effect before scale (5); tiers and the external benchmark (6) last. A comparison’s outcome changes what is built next; the contract does not change.

6 Results

Reserved. Nothing here has been trained; the tables fix the columns now. Each cell will carry its seed count, its paired interval and the label retrospective or as-issued.

6.1 The forecasting benchmark

arm	SST, MSSS _{damped}			heat content 0–700 dbar		runtime
	1 mo	3 mo	6 mo	3 mo	6 mo	
B1 damped persistence	0	0	0	0	0	—
B1 LIM, data space (EOF)	—	—	—	—	—	—
B2 direct multi-lead regression, history	—	—	—	—	—	—
B3 fixed codec + embedding-space LIM	—	—	—	—	—	—
B4 B3 + small nonlinear residual	—	—	—	—	—	—

Table 5: Comparison (0), decoded physical space, identical origins, history and targets, paired intervals on the differences between arms; the same table for ACC, amplitude and bias, and against the evolving climatology.

quantity	score	1 mo	3 mo	6 mo	12 mo
Niño-3.4 index (exact integral)	ACC / RMSE	—	—	—	—
subpolar-gyre SST	ACC / RMSE	—	—	—	—
tropical Indian Ocean SST	ACC / RMSE	—	—	—	—
heat content 0–700 dbar, Argo-covered	ACC / RMSE	—	—	—	—
El Niño onset	threshold probability, duration, accumulated anomaly	—	—	—	—

Table 6: The decoded panel of §5.1, reported for every arm of every comparison beside the dense-field score, each cell against the stationary and the evolving climatology.

6.2 Encoder: hourglass against snapshot

codec	one-step ratio	lead 1 / bar	lead 3 / bar	lead 6 / bar	T1–T2
snapshot (waist only)	—	—	—	—	—
hourglass, fixed (A0)	—	—	—	—	—

Table 7: Comparison (1). Ratios: held-out loss over the predict-the-mean bar; one-step ratio: the forecaster’s z -space error over persistence.

6.3 Cone geometry

arm	R1 primary	R1 lead 1	R1 lead 2	R1 lead 3	R2	R3	tokens
A0 fixed	0 (reference)	—	—	—	—	0 by construction	—
A1 per channel	—	—	—	—	—	—	—
A1g per group	—	—	—	—	—	—	—
A2 primed	—	—	—	—	—	—	—
A1 without ring	—	—	—	—	—	—	—
A1, coarse global summary for ring	—	—	—	—	—	—	—

Table 8: Comparison (2), three seeds per arm, paired, equal token budget, one fixed target set. R1 primary is the difference from A0 on the primary score with its paired interval; R3 is descriptive. R4 is a figure.

6.4 Forecaster and the null ladder

lead (pentads)	1	3	8	21	34	55	73
persistence	—	—	—	—	—	—	—
damped persistence	—	—	—	—	—	—	—
LIM, data space	—	—	—	—	—	—	—
LIM, embedding space (the backbone alone)	—	—	—	—	—	—	—
direct multi-lead regression	—	—	—	—	—	—	—
analogue	—	—	—	—	—	—	—
forecaster, backbone + residual	—	—	—	—	—	—	—
with fine-tuned $\mathbf{L}$	—	—	—	—	—	—	—
without backbone	—	—	—	—	—	—	—
one-step, rolled	—	—	—	—	—	—	—
dense lags	—	—	—	—	—	—	—

Table 9: Comparison (3) and the null ladder: $\text{MSSS}_{\text{clim}}$ on the dense field, world ocean, area-weighted, per lead; the same table for $\text{MSSS}_{\text{damped}}$, ACC, amplitude and bias, per channel and scope, and the backbone’s $\max_{\ell} \|\mathbf{G}_{\ell}\|_2$ beside its row.

6.5 Observation stores per read-out

read-out	score	null	without stores	with stores
El Niño	Niño-3.4 correlation at six months	warm-water-volume persistence	—	—
carbon sink	held-out surface- CO_2 error	climatological null	—	—
overturning	velocity loss / bar; transport r at 26.5°N	drifter-free cone; damped persistence	—	—
SST	level-3 cell loss / bar	drifter- and ship-free cone	—	—
land biosphere	tower flux error / bar; canopy height error at laser footprints	site climatology, damped persistence; the 2020 map, previous value	—	—

Table 10: Comparison (4): the stores per read-out, and the footprint fields, against the same cone without them. The biosphere row is also scored with AlphaEarth’s cache in place of the tile codec.

6.6 Scaling

width	held-out minimum step	one-step ratio	mean ACC, leads 1–73	mean $\text{MSSS}_{\text{damped}}$
—	—	—	—	—

Table 11: Comparison (5): the width ladder at L2, one row per width.

6.7 Weather against FGN

channel	lead	fair CRPS			ensemble-mean RMSE		
		FGN	ours	damped	FGN	ours	damped
2 m temperature	1	—	—	—	—	—	—
	2	—	—	—	—	—	—
	3	—	—	—	—	—	—
10 m wind speed	1	—	—	—	—	—	—
	2	—	—	—	—	—	—
	3	—	—	—	—	—	—
surface pressure	1	—	—	—	—	—	—
	2	—	—	—	—	—	—
	3	—	—	—	—	—	—
precipitation	1	—	—	—	—	—	—
	2	—	—	—	—	—	—
	3	—	—	—	—	—	—

Table 12: Comparison (6): the atmospheric channels on 2023, pentad means on the 1° cells, leads in pentads (5, 10, 15 days). FGN: the released WeatherNext 2 checkpoints from the operational analysis, its four released seeds pooled into one ensemble of at least fifty members; ours: the seed ensemble of §2.6; damped: per-cell AR(1). A system benchmark with unmatched input information (§5.1). The same table for spread-skill ratio and 10° -pooled CRPS, and the arms with and without the fine stores.

7 Related work

Global data-driven weather models [24, 25, 27, 26] and their hybrids with a dynamical core [28], atmospheric foundation models [29, 30], end-to-end systems reading observations [31], ocean emulators [32, 33] and coupled emulators [34] read one grid, one variable set and one reach. This design reads every source at its own footprint through one token, samples a per-channel cone, and predicts through a query carrying the footprint of the answer. Speed times time as a receptive field has precedents: an ocean emulator drops lateral boundaries because eddies travel a bounded distance over the horizon [33]; a neural semi-Lagrangian layer gathers at $\mathbf{x} - \Delta t \mathbf{u}$ with one velocity shared across transported modes [35]; deformable attention learns offsets per query and head with a learning-rate multiplier [8]. Truncation by memory, a cone per channel, and a learnable ellipse with a hierarchical per-channel drift are, to the authors’ knowledge, not stated as a principle elsewhere; per-variable tokenisation [29, 30] changes what a variable is embedded as, not where it is read from. The architecture is a masked autoencoder [9] with Perceiver cross-attention [10]; AlphaEarth Foundations [2] is the closest embedding-field model, over land at annual cadence, and enters here as a token cache; geospatial tile models [36] embed imagery. Global canopy-height maps [46, 47, 48] regress one year of laser footprints from imagery, one map per year; here the footprints are targets in their own coordinates and the maps are one more target, so the forecast of a canopy is scored against the laser, not against a map. Frozen backbones with shallow read-outs [37] are the evaluation stance, judged by what a forecaster predicts from the representation rather than by a current-state probe. The LIM [20, 21] and the interannual-to-decadal verification framework [22] are the nulls and scores, with Murphy’s decomposition [13] as the calibration check; the decoupled objective is the faithful heteroscedastic regression of [12], distinct from β -NLL [11]; the fair CRPS is Ferro’s estimator [14, 15, 16] within the proper scoring rules of [23]; the four-epoch rule [17] and the compute-optimal ratio [18] are planning heuristics; sparse experts [19]. The physics of the per-channel carriers is from [3, 4, 5, 6]; the overturning array [38] and Argo [39] enter as observations in their own coordinates.

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A Data and code availability

Code: <https://github.com/blauewelt/earth> — geometry (ml/cone.py), sampler (ml/cone_sampler.py), codec and masking plan (ml/cone_codec.py, ml/train_cone.py), point-store reader (ml/family10_store.py), builders (ml/build_family7.py, ml/build_family8_argo.py, ml/build_family10_stores.py, ml/build_family1_stores.py), registry writers (ml/build_family10_registry.py, ml/build_family1_registry.py). Data, in the Hugging Face dataset `chfrank/earth-tensors` under `tensors/`: the 0.25° pentad grid (`family7_global025_pentad_l2/manifest.json`), the Argo store (`family8_argo_l0/`), and four registries of checksums, sources, licences and build state — `family10_2/family10.json` (the grid’s groups and the six point stores, with the `siblings` block), `family1_gf/family1gf.json`, `family1_tf/family1tf.json` and `family09_tf/family09tf.json`; private-track stores are listed there and held in `chfrank/earth-tensors-private`. Handovers, each self-contained and measured against the published bytes: `docs/FAMILY7_DATA_HANOVER.md`, `docs/FAMILY71_INGEST_HANOVER.md`, `docs/FAMILY8_DATA_HANOVER.md`, `docs/FAMILY10_DATA_HANOVER.md`, `docs/FAMILY1GF_DATA_HANOVER.md`, `docs/FAMILY1TF_DATA_HANOVER.md`, `docs/FAMILY09TF_DATA_HANOVER.md`. Design notes, one per family, in `ml/paper/notes/`: `family7.tex`,

family71.tex, family72.tex, family10.tex, family1gf.tex, family1tf.tex, family09tf.tex;
plans and verdict rules: ml/plans/. Sources: GLORYS12 and GREP (Copernicus Marine), OISST
v2.1 (NOAA NCEI), NCEP/NCAR Reanalysis-1 (NOAA PSL), the Roemmich–Gilson Argo climatology
(Scripps; Argo data from the International Argo Program), OC-CCI v6.0 (ESA CCI, PML), RAPID-
MOCHA-WBTS (NERC, NSF, NOAA), the Global Drifter Program and GTMBA (NOAA AOML,
PMEL), SOCAT, Copernicus Marine along-track sea level, Global Fishing Watch apparent fishing effort
v3.0 (CC BY-NC 4.0), GEDI (NASA LP DAAC, ORNL DAAC), ICESat-2 ATL08 (NSIDC), TROPOMI
fluorescence (Caltech, ESA), FLUXNET, GBIF, MODIS and VIIRS land products (LP DAAC), GLAD,
ETH and Meta canopy height, ESA CCI Biomass, Hansen GFC, PACE (NASA OB.DAAC), and the
sources named per store in the registries.